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Comparative effects of first-touch shooting and shooting after dribble training on shooting accuracy in youth soccer players

Efectos comparativos del entrenamiento de remate al primer toque frente al remate tras conducción en la precisión de tiro de futbolistas jóvenes

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Abstract
In soccer, shooting opportunities may arise either immediately after receiving the ball or after dribbling, making both situations relevant to shooting practice in youth players. This study compared the effects of First-Touch Shooting and Shooting After Dribble training on shooting accuracy in youth soccer players. An experimental study using a two-group pretest–posttest design involved 18 youth soccer players, allocated to two groups (n = 9 per group) using matched allocation. One group performed First-Touch Shooting drills, whereas the other performed Shooting After Dribble drills. Both groups completed 12 training sessions, conducted three times per week. Shooting accuracy was assessed using the Bobby Charlton Shooting Test. Data were analyzed using paired-samples t-tests and analysis of covariance (ANCOVA), with baseline shooting accuracy included as a covariate. Effect sizes were calculated using Cohen's d, with statistical significance set at $p < 0.05$. The First-Touch Shooting group showed a statistically significant pretest–posttest improvement ($p = 0.028$; $d = -0.891$), as did the Shooting After Dribble group ($p = 0.005$; $d = -1.273$). However, ANCOVA detected no statistically significant between-group difference in posttest shooting accuracy after adjustment for baseline performance ($F = 0.994$; $p = 0.335$). These findings indicate that shooting accuracy improved within both groups during the intervention period, but the analysis did not establish a statistically significant difference between the two training methods. The results should not be interpreted as evidence that the methods are equivalent, particularly given the small sample size and the absence of a non-training control group.

Keywords: first-touch shooting; shooting after dribble; training methods; shooting accuracy

Resumen
En el fútbol, las oportunidades de tiro pueden surgir inmediatamente después de recibir el balón o tras realizar un regate; por ello, ambas situaciones son relevantes para el entrenamiento del remate en jugadores jóvenes. Este estudio comparó los efectos del entrenamiento de tiro al primer toque frente al entrenamiento de tiro tras regate en la precisión de tiro de futbolistas jóvenes. Se llevó a cabo un estudio experimental con un diseño de dos grupos (pretest-posttest) en el que participaron 18 futbolistas jóvenes, distribuidos en dos grupos (n = 9 por grupo) mediante una asignación equilibrada. Un grupo realizó ejercicios de tiro al primer toque, mientras que el otro llevó a cabo ejercicios de tiro tras regate. Ambos grupos completaron 12 sesiones de entrenamiento, a razón de tres sesiones por semana. La precisión de tiro se evaluó mediante la prueba de tiro «Bobby Charlton» (Bobby Charlton Shooting Test). Los datos se analizaron utilizando pruebas t para muestras emparejadas y un análisis de covarianza (ANCOVA), incluyendo la precisión de tiro inicial como covariable. El tamaño del efecto se calculó mediante la d de Cohen, estableciéndose la significación estadística en $p < 0,05$. El grupo de tiro al primer toque mostró una mejora estadísticamente significativa entre el pretest y el posttest ($p = 0,028$; $d = -0,891$), al igual que el grupo de tiro tras regate ($p = 0,005$; $d = -1,273$). Sin embargo, el ANCOVA no detectó diferencias estadísticamente significativas entre ambos grupos en la precisión de tiro tras el posttest, una vez realizado el ajuste según el rendimiento inicial ($F = 0,994$; $p = 0,335$). Estos hallazgos indican que la precisión de tiro mejoró en ambos grupos durante el periodo de intervención; no obstante, el análisis no reveló

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una diferencia estadísticamente significativa entre los dos métodos de entrenamiento. Estos resultados no deben interpretarse como una prueba de que ambos métodos sean equivalentes, especialmente dado el reducido tamaño de la muestra y la ausencia de un grupo de control que no realizara entrenamiento.

Palabras clave: tiro al primer toque; tiro tras conducción; métodos de entrenamiento; precisión de tiro

Introduction

The ability to convert chances into goals is one of the key indicators of offensive performance in modern soccer (Anzer & Bauer, 2021; Gonzalez-Rodenas et al., 2020). Even if a team creates many chances, the outcome of a match is still determined by the effectiveness of finishing through accurate shooting (Lamas et al., 2020; Tenga et al., 2010). Biomechanical analysis shows that shooting is the most critical technical skill in the goal-scoring process because it requires precise coordination of movement among body segments, ball orientation, and the target (Lees et al., 2010; Shan, 2022; Shan & Zhang, 2011). Developing shooting accuracy is a top priority in youth player development because this skill directly contributes to the effectiveness of the attack and the team's overall performance (Paisal et al., 2024). In addition to technical training, physical readiness also plays an important role in youth soccer performance. Previous research demonstrated that the FIFA 11+ Kids warm-up programme significantly improved agility in youth soccer players, highlighting that physical preparation can support the execution of soccer-specific technical skills (Anam, Setiowati, Indardi, Irawan, Aditia, et al., 2024; Hanief et al., 2025).

Shooting success depends not only on the quality of the shot but also on the player's ability to integrate various skill components in dynamic game situations. During a match, players receive passes, control the ball, adjust their body orientation, identify opponents' and the goalkeeper's positions, and then decide whether to take a shot (Sarifudin et al., 2023). This process demonstrates that shooting accuracy results from the simultaneous interaction among ball control, balance, coordination, visual perception, and decision-making (Shan, 2022). Training that focuses solely on kicking technique without incorporating game context tends to

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result in low skill transfer when applied in match situations (Browne et al., 2020; Pinder et al., 2011).

Research on skill acquisition indicates that training effectiveness is influenced by the alignment between training characteristics and the performance demands athletes face during competition (Bergmann et al., 2021). The principle of training specificity states that skill adaptation develops optimally when training closely matches the characteristics of the actual task (Stone et al., 2022). The concept of Representative Learning Design reinforces this view by emphasizing the connection between perception, decision-making, and movement execution during training, enabling the acquired skills to be more easily transferred to game situations (Petiot et al., 2024). Recent systematic reviews also indicate that training designs that maintain the game context result in more effective skill learning than technical drills performed in isolation (Bergmann et al., 2021).

Evidence from professional soccer matches further underscores the importance of finishing immediately after receiving the ball. An analysis of more than 2,000 shooting attempts shows that the majority of goals result from first-touch shots, that is, finishing performed immediately after receiving the ball without maintaining possession for an extended period (Perez-Arroniz et al., 2025). This characteristic reduces the reaction time of both defenders and goalkeepers, thereby increasing the likelihood of a goal. Although the effectiveness of first-touch finishing has been demonstrated through match analysis, research that translates this skill into structured training interventions remains relatively limited.

Various training exercises have been developed to improve shooting accuracy in youth soccer players (Ben Kahla et al., 2022). These findings indicate that different training variations can improve shooting performance. However, previous studies have evaluated each method separately or compared them to other training models with different characteristics. Consequently, the relative effectiveness of training that emphasizes first-touch control before shooting versus training that emphasizes dribbling before shooting has not yet been comprehensively explained.

In soccer, shooting accuracy may depend not only on kicking technique but also on the actions preceding the shot, including receiving and controlling a pass or dribbling the ball (Cao,

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2024; Sarmento et al., 2018). First-Touch Shooting emphasizes receiving a pass, controlling the ball with the first touch, adjusting body orientation, and shooting in sequence. In contrast, Shooting After Dribble requires players to maintain ball control while dribbling before executing a shot. These approaches represent different technical sequences that may be incorporated into youth soccer shooting practice.

Initial observations at the soccer academy indicated that some players had difficulty maintaining shooting accuracy after receiving a pass or dribbling. These observations suggest the need to examine training approaches that address shooting accuracy in different technical situations. Previous research has reported that mastery of basic soccer techniques among players aged 13–15 years remains moderate, supporting the importance of structured and appropriately designed skill training (Hamid et al., 2024).

Although different shooting drills are used in soccer training, further empirical comparison of First-Touch Shooting and Shooting After Dribble under the same training and assessment conditions may help clarify their relative outcomes. Therefore, this study aimed to compare changes in shooting accuracy following First-Touch Shooting and Shooting After Dribble training in youth soccer players using a two-group pretest–posttest design.

Materials And Methods

Participants

Participants were recruited from a single soccer academy using a feasibility-based convenience sampling approach. The sample included 18 eligible male youth soccer players registered in the academy's U-13 age category. Participants were 10–13 years old at the time of data collection. The sample size was determined by the number of eligible players available at the academy during the study period; no formal a priori sample size calculation was performed. The inclusion criteria were: (1) players registered in the under-13 age category, (2) players who attended practice regularly, (3) players who were healthy and injury-free during the study period, and (4) players who were willing to participate in the entire study.

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TABLE 1. Participant Characteristics (n = 18)

Variable	Mean ± SD	Minimum	Maximum
Age (years)	11.17 ± 0.92	10	13
Height (cm)	139.89 ± 7.45	131	156
Weight (kg)	34.39 ± 8.93	24	54
Body Mass Index (kg/m²)	17.36 ± 3.12	13.80	24.70

This study was conducted in accordance with the Declaration of Helsinki. Ethical approval was obtained from the Health Research Ethics Committee, Faculty of Medicine, Universitas Negeri Semarang (No. 331/KEPK/FK/KLE/2026). Written informed consent was obtained from the parents or legal guardians of all participants before participation.

Methods

This study used a two-group pretest–posttest experimental design to compare the effectiveness of the First-Touch Shooting and Shooting After Dribble training methods in improving youth soccer players' shooting accuracy.

Before the intervention, all participants completed a pretest to assess their baseline shooting accuracy. The pretest scores were ranked in descending order, from the highest to the lowest, and participants were allocated to the two training groups using the ordinal pairing technique with a repeating A–B–B–A pattern. This procedure continued until all 18 participants had been assigned, with nine participants in each group, to help distribute baseline shooting ability more evenly. Group A received the First-Touch Shooting intervention, whereas Group B received the Shooting After Dribble intervention. Group allocation was performed by an individual who was not involved in outcome assessment. The assessor conducting the pretest and posttest assessments was blinded to the participants' group assignments to minimize potential assessment bias.

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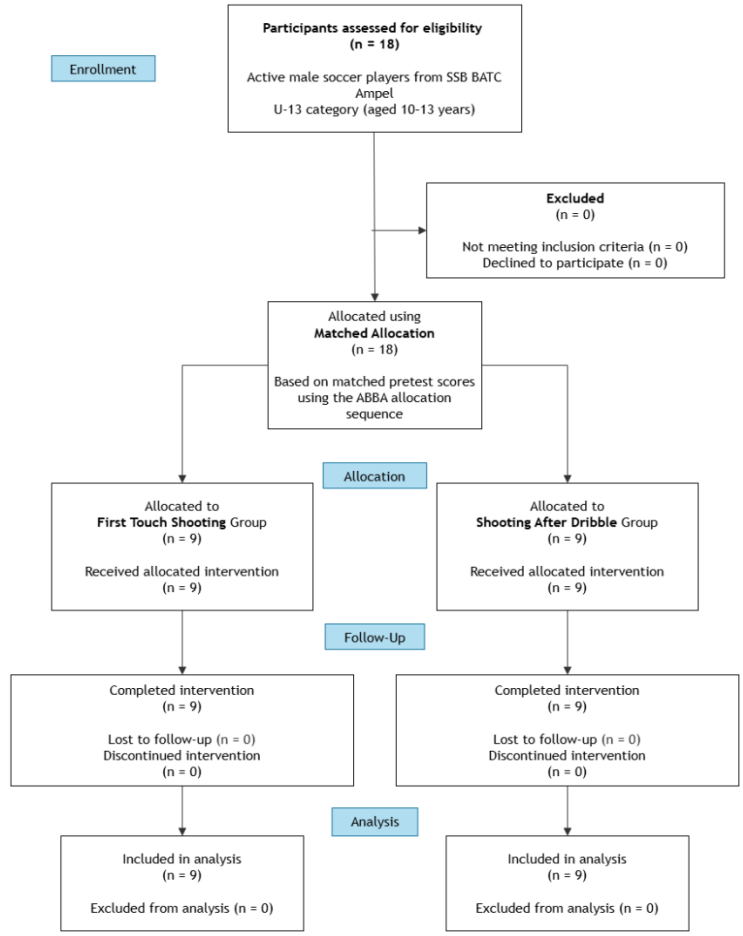


FIGURE 1. Flow diagram of participant recruitment, matched allocation, follow-up, and analysis.

Procedure

In the initial stage, all participants completed a pretest using the Bobby Charlton Shooting Test to obtain baseline shooting accuracy scores. Each participant performed three shooting trials, and the scores were summed to obtain the final pretest score. Following the pretest, each group participated in a 12-session training program conducted three times per week. Each training session lasted approximately 60–90 minutes and included a warm-up, core exercises, game-based training in some sessions, and a cool-down. The training program progressed gradually through changes in movement complexity, training volume, and the application of game situations. After completing the intervention, all participants completed a posttest using the same assessment instrument and procedures as the pretest.

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During the intervention period, both groups trained at the same location and followed the same schedule, training frequency, session duration, instructions, rest periods, and training facilities under the supervision of the same coach. These arrangements were intended to minimize the influence of external variables. The main difference between the groups was the specific shooting exercise performed: First-Touch Shooting or Shooting After Dribble.

TABLE 2. Characteristics of the Intervention Program

Component	First-Touch Shooting Group	Shooting After Dribble Group
Number of participants	9 players	9 players
Number of sessions	12 sessions	12 sessions
Frequency	3 times a week	3 times a week
Duration of training	Same	Same
Training location	Same	Same
Trainer	Same	Same
Focus of the intervention	*First touch → shooting*	*Dribbling → shooting*

Training Intervention

The intervention consisted of 12 training sessions conducted three times per week. Both groups participated in structured training sessions comprising preparation and warm-up, the assigned shooting intervention, and cool-down activities. The main training component varied in duration and volume across sessions. As documented in the training program, the prescribed volume included 10 repetitions, 2 sets of 10 repetitions, or 3 sets of 10 repetitions, depending on the session. General rest periods of 1–2 minutes were scheduled between training components, rather than as standardized rest intervals between sets.

The First-Touch Shooting group practiced receiving a pass, controlling the ball with the first touch toward the available shooting space, and subsequently shooting at the goal. The drills included passing sequences and movements to receive the ball before shooting. The Shooting After Dribble group practiced dribbling toward the goal, including zigzag dribbling through

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four markers and performing a feint to pass an additional marker before shooting. The documented dribbling distance was approximately 10 m in several sessions, although some sessions used different distances or drill configurations.

Selected sessions also included small-sided or game-based activities to encourage the application of the practiced techniques in playing situations. The training log documented variations in game format, rules, and duration across sessions. Each group received its assigned training intervention throughout the program.

First-Touch Shooting Intervention

The First-Touch Shooting drill focuses on players' ability to receive a pass, control the ball with their first touch, direct the ball into the shooting lane, and then immediately take a shot at the goal. Throughout the program, the drills progressed from basic technical practice to more complex shooting exercises. The intervention focused on practicing ball reception, first-touch control, body orientation, and shooting accuracy.

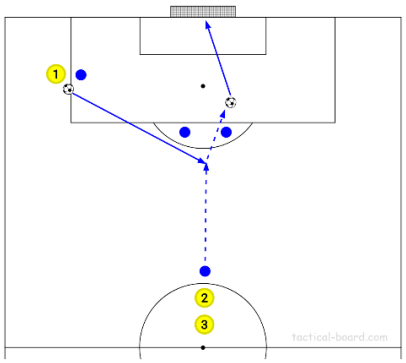


FIGURE 2. Diagram of the First-Touch Shooting Drill

Shooting After Dribble Intervention

The Shooting After Dribble drill involved controlled dribbling, changes of direction, feints, and shooting toward the goal. The intervention focused on practicing ball control and shooting accuracy following dribbling.

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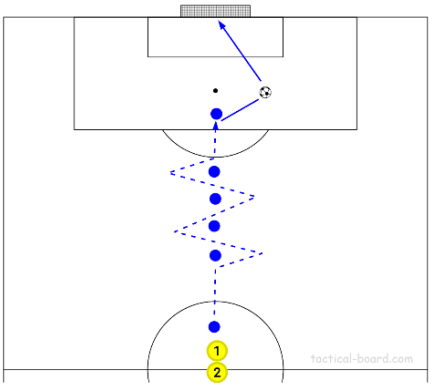


FIGURE 3. Diagram of the Shooting After Dribble Drill

After the entire training program was completed, all participants took a posttest using the same assessment instruments and procedures as the pretest to ensure consistency in the assessment results.

Shooting accuracy was assessed using the Bobby Charlton Shooting Test (Mielke, 2003). The test involved shooting toward a goal divided into six scoring zones, as illustrated in Figure 4. The target measured 7.32 m in width and 2.44 m in height, with the shooting point positioned 16 m from the goal. Each participant performed three shooting trials, and the scores from all three trials were summed to produce the final shooting accuracy score, with a maximum possible score of 150 points. The scoring system assigned different point values to the target zones (10, 20, 40, and 50 points), as shown in Figure 4. The same test instrument, shooting distance, scoring system, and testing procedure were used for both the pretest and posttest to maintain consistency in outcome measurement. Although several international shooting assessments have been developed and validated (Ali et al., 2007; Engler et al., 2023; Russell et al., 2010), the Bobby Charlton Shooting Test was selected as a practical field-based measure of shooting accuracy in youth soccer players.

The test was selected because its standardized shooting distance and zone-based scoring system provided a practical means of quantifying shooting accuracy under consistent field-testing conditions. However, the present study did not independently establish the validity or reliability of this test specifically for players aged 10–13 years.

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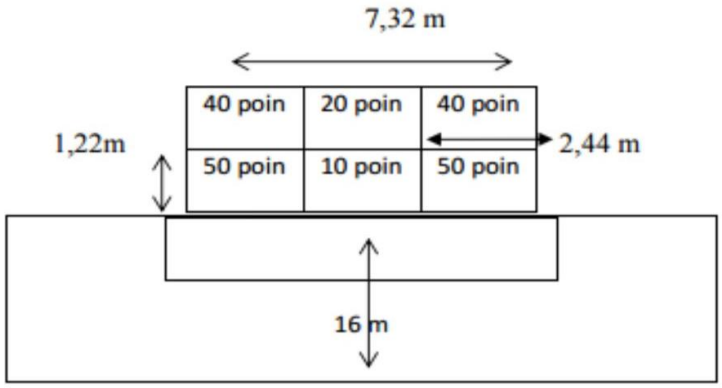


FIGURE 4. Score Distribution on the Bobby Charlton Shooting Test

Data analysis

Data analysis was performed using IBM SPSS Statistics version 27 for all primary statistical analyses, while JASP version 0.98.1 was used to calculate effect sizes and 95% confidence intervals (95% CIs). Descriptive statistics were used to describe participants' characteristics and measurement results, including means, standard deviations, minimums, and maximums.

Before testing the hypotheses, statistical assumptions were tested. Data normality was assessed using the Shapiro–Wilk test because the number of participants per group was less than 50, making this test more suitable for small samples (Mishra et al., 2019). The homogeneity of variances across groups was evaluated using Levene's test. In contrast, the assumption of homogeneity of regression slopes was tested to ensure that the relationship between pretest scores (covariates) and posttest scores (dependent variable) was consistent across all groups. The fulfillment of these assumptions formed the basis for using Analysis of Covariance (ANCOVA) as the primary analysis (Schwarz, 2025).

Changes in shooting accuracy within each group were analyzed using paired-samples t-tests by comparing pretest and posttest scores. Differences between the First-Touch Shooting and Shooting After Dribble training groups were then examined using analysis of covariance (ANCOVA), with pretest shooting accuracy as the covariate, posttest shooting accuracy as the dependent variable, and treatment group as the fixed factor. This approach was used to account for participants' baseline shooting accuracy when comparing posttest outcomes.

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In addition to statistical significance, this study reports effect sizes to describe the magnitude of the observed changes. Within-group effect sizes were calculated using Cohen's d and are presented with 95% confidence intervals (CIs). The between-group effect size for the ANCOVA was reported using partial eta squared (η^2p). All statistical tests were conducted at a significance level of $\alpha = 0.05$.

Results

A total of 18 youth soccer players completed the study, with nine participants in the First-Touch Shooting group and nine in the Shooting After Dribble group. The baseline characteristics of the participants are presented in Table 3.

TABLE 3. Baseline Characteristics of Participants

Variable	First-Touch Shooting (n = 9)	Shooting After Dribble (n = 9)	p
Age (years)	11.11 ± 0.93	11.22 ± 0.97	0.807
Height (cm)	137.89 ± 6.33	141.89 ± 8.30	0.267
Weight (kg)	33.22 ± 7.33	35.56 ± 10.62	0.595
Body Mass Index (kg/m²)	17.36 ± 2.88	17.36 ± 3.53	1.000

The baseline characteristics of the two groups did not differ significantly across all observed variables ($p > 0.05$). Thus, both groups had comparable baseline conditions prior to the intervention. Before conducting inferential analysis, the data were first tested to ensure they met the assumptions of parametric analysis. The results of the normality test using the Shapiro–Wilk test, the homogeneity test using Levene's test, and the test for homogeneity of regression slopes are presented in Table 4.

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TABLE 4. Results of the parametric assumption tests

Testing	Statistics	Values	Variables	*p*
Shapiro–Wilk	W	0.948	Pretest First-Touch Shooting	0.663
	W	0.967	Posttest First-Touch Shooting	0.215
	W	0.893	Pretest Shooting After Dribble	0.869
	W	0.956	Posttest Shooting After Dribble	0.752
Levene's Test	F	0.492	Both groups	0.493
Homogeneity of regression slopes	F	0.140	Group × pretest interaction	0.714

NOTE: The statistical values show the W value from the Shapiro–Wilk test and the F value from Levene's test, as well as the homogeneity of regression slopes.

No statistically significant departures from normality were detected by the Shapiro–Wilk tests, although these tests have limited power with small samples. Levene's test did not indicate unequal variances, and the group × pretest interaction was not statistically significant, supporting the homogeneity-of-regression-slopes assumption. Therefore, the paired-samples t-test and analysis of covariance (ANCOVA) were used for the planned analyses.

TABLE 5. Changes in shooting accuracy scores for each group

Groups	Pretest (Mean ± SD)	Posttest (Mean ± SD)	Mean increase	t	p	Cohen's d	SE Cohen's d	95% CI for Cohen's d
First-Touch Shooting	73.33 ± 33.54	90.00 ± 36.74	16.67	-2.673	0.028	-0.891	0.207	−1.654 to −0.090
Shooting After Dribble	76.67 ± 32.40	102.22 ± 33.08	25.55	-3.820	0.005	-1.273	0.275	−2.149 to −0.357

NOTE: Paired-samples t-tests were conducted using the pretest and posttest scores. The effect size and its 95% confidence interval follow the JASP output convention, with the difference calculated as pretest minus posttest. Negative Cohen's d values therefore indicate higher posttest scores than pretest scores.

The paired-samples t-test showed statistically significant increases in shooting accuracy in both groups. The First-Touch Shooting group improved from 73.33 ± 33.54 at pretest to 90.00

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± 36.74 at posttest (t(8) = −2.673, p = 0.028, Cohen's d = −0.891, 95% CI [−1.654, −0.090]). The Shooting After Dribble group improved from 76.67 ± 32.40 to 102.22 ± 33.08 (t(8) = −3.820, p = 0.005, Cohen's d = −1.273, 95% CI [−2.149, −0.357]). The absolute effect-size magnitudes were large in both groups. These within-group findings indicate improvements over the intervention period but do not establish that either training method was superior to the other.

A comparison of the two training methods was conducted using analysis of covariance (ANCOVA), with pretest scores as a covariate. The results are presented in Table 6.

TABLE 6. Results of the Analysis of Covariance (ANCOVA)

Source of Variation	**df**	**F**	**p**	**Partial η²**
Corrected Model	2	18.623	<0.001	0.713
Pretest	1	35.510	<0.001	0.703
Group	1	0.994	0.335	0.062
Error (Residual)	15	-	-	-

NOTE: Posttest scores were the dependent variable, with pretest scores included as a covariate.

Pretest scores had a significant effect on posttest scores. However, after controlling for the effect of initial ability, no significant difference was found between the First-Touch Shooting group and the Shooting After Dribble group.

To supplement the ANCOVA results, adjusted posttest means were calculated, representing the average posttest scores after correction for pretest scores. The results of these calculations are presented in Table 7.

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TABLE 7. Adjusted posttest means and between-group comparison based on the ANCOVA model

Groups / comparison	Adjusted posttest mean	95% CI
First-Touch Shooting	91.574	77.065–106.083
Shooting After Dribble	100.840	86.330–115.350
Adjusted mean difference (Shooting After Dribble − First-Touch Shooting)	9.266	−11.254 to 29.785

NOTE: Adjusted means were estimated at a pretest score of 75.00. The adjusted mean difference was based on pairwise comparisons of estimated marginal means with Bonferroni adjustment (p = 0.349).

The adjusted posttest mean was 100.840 points in the Shooting After Dribble group and 91.574 points in the First-Touch Shooting group. The adjusted mean difference was 9.266 points (95% CI: −11.254 to 29.785; Bonferroni-adjusted p = 0.349). The ANCOVA showed no statistically significant between-group difference after adjustment for baseline shooting accuracy ($F(1, 15) = 0.994$, $p = 0.335$, partial $\eta^2 = 0.062$). Although the adjusted mean was numerically higher in the Shooting After Dribble group, the results do not establish that either training method was superior or that the two methods were equivalent.

Discussion

Both the First-Touch Shooting and Shooting After Dribble groups showed improvements in shooting accuracy following the intervention. However, the ANCOVA results indicated that the between-group difference was not statistically significant after adjustment for baseline shooting accuracy. Although the Shooting After Dribble group showed a numerically higher adjusted posttest mean and a larger within-group effect size, these findings should be interpreted cautiously because of the small sample size. The results therefore indicate improvements within both groups but do not establish the superiority or equivalence of either training method. When interpreting intervention outcomes, both statistical significance and effect sizes should be considered (Williams & Hodges, 2023).

The observed pattern of results may be interpreted in light of the similar task demands of the two training methods. Both First-Touch Shooting and Shooting After Dribble require

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players to control the ball, adjust their body orientation toward the target, maintain balance, position the supporting foot, and execute the shot accurately. These shared task characteristics may help explain why shooting accuracy improved in both groups. However, the present study did not establish that the training methods produce equivalent improvements. Further research is needed to determine how differences in the technical sequences influence shooting performance (Shan, 2022; Stone et al., 2022).

These findings can be discussed in relation to previous research on task-specific skill training. Bergmann et al. (2021) emphasized the importance of maintaining relevant perceptual information during skill learning, while Choo et al. (2024) highlighted the value of training designs that reflect the characteristics of the performance task (Bergmann et al., 2021; Choo et al., 2024). The principles of training specificity, the Constraints-led Approach, and Representative Learning Design provide theoretical perspectives for understanding how task characteristics and perception–action coupling may relate to sport-specific skill development. However, the present study did not directly assess training design quality, perception–action coupling, or the independent contribution of these factors to shooting accuracy. Therefore, these perspectives should be considered possible explanations rather than established mechanisms underlying the observed improvements.

In addition to being influenced by training design, performance improvement is also related to players' ability to utilize information during training. Each shooting attempt requires players to recognize the ball's position, estimate the shooting space, adjust their body orientation, and determine the timing of contact before releasing the shot. This continuous interplay between perception and action enables players to generate increasingly efficient and adaptive motor responses to situational demands (Bergmann et al., 2021; Choo et al., 2024). This may offer one possible theoretical explanation for the improvements observed in both groups, although the present study did not directly test the mechanisms underlying these changes. These findings are also consistent with previous evidence showing that better functional movement quality is associated with improved physical function and lower movement-related injury risk, highlighting the importance of developing movement

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competency as a foundation for sport-specific skill acquisition (Anam et al., 2026; Anam, Setiowati, Indardi, Irawan, Pavlović, et al., 2024).

As training experience increases, players become increasingly capable of selecting movement solutions appropriate to the conditions they face—whether in determining body position, regulating shot power, or utilizing available shooting space. This adaptation process is part of motor learning, which refers to relatively permanent changes in motor ability resulting from training experience (Leech et al., 2022). From a modern motor learning perspective, performance improvement is demonstrated not only by increased movement precision but also by improved coordination efficiency, performance consistency, and the ability to maintain movement quality across varying situations (Williams & Hodges, 2023).

Overall, shooting accuracy improved in both training groups following the intervention. The findings suggest that both training approaches may be useful for practicing shooting skills in youth soccer players. However, because the between-group difference was not statistically significant and the sample was small, the results do not establish that the methods have equivalent effects. The potential role of task characteristics and instructional design in explaining the observed improvements should therefore be considered a theoretical interpretation that requires further investigation.

Although the Shooting After Dribble group showed a numerically higher adjusted posttest mean and a larger within-group effect size than the First-Touch Shooting group, the ANCOVA did not detect a statistically significant between-group difference after adjustment for baseline shooting accuracy. The numerical differences may be relevant for generating hypotheses about the practical effects of the training methods, but they should not be interpreted as evidence of superiority or equivalence. Future studies with larger samples and appropriate precision estimates are needed to clarify the magnitude and practical relevance of the differences between the two approaches.

The results of this study reinforce those of Anam et al., who reported that systematic, goal-oriented shooting drills can improve soccer players' shooting accuracy (Anam et al., 2021). However, that study evaluated only one type of drill and therefore could not explain the relative effectiveness of different drill variations. Similar findings showed that Shooting After

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Dribble training improves shooting accuracy (Fernando H. Girsang & Supriadi, 2021). Nevertheless, this study indicates that the descriptive advantage of Shooting After Dribble does not translate into a significant difference after controlling for players' initial ability. One possible theoretical interpretation is that shared task characteristics may contribute to improvements in shooting accuracy across different drill formats. However, the present study did not directly test the independent contribution of training design quality or movement complexity to performance improvement.

Research by Paisal et al. also reported improvements in shooting accuracy through various technical training exercises (Paisal et al., 2024). However, most of these studies evaluated each method separately, using pre- and post-training analyses. This approach demonstrates that training effectively improves performance but fails to explain whether one method is superior to another after accounting for participants' initial ability. This study provides a comparative perspective on two shooting-training variations with similar task characteristics. However, the non-significant between-group difference should not be interpreted as evidence that the two methods produce equivalent improvements.

This study contributes to the literature by directly comparing First-Touch Shooting and Shooting After Dribble training in youth soccer players while using analysis of covariance (ANCOVA) to account for baseline shooting accuracy. The findings showed improvements in shooting accuracy in both training groups; however, no statistically significant between-group difference was found after adjustment for baseline performance. Although the Shooting After Dribble group showed a numerically higher adjusted posttest mean, this result should be interpreted cautiously given the small sample size. Therefore, the findings provide preliminary comparative evidence regarding these two training approaches but do not establish that they are equivalent in effectiveness. Further studies involving larger samples and more rigorous experimental designs are needed to clarify their comparative effects on shooting accuracy in youth soccer players.

The two methods may be selected as complementary training alternatives according to the objectives of a training session. First-Touch Shooting may be used to practice shooting after receiving a pass, whereas Shooting After Dribble may be used to practice shooting following

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ball progression through dribbling. The present findings do not establish the superiority of either method.

This study has several limitations that should be noted. The relatively small sample size, drawn from a single soccer academy, limits the generalizability of the study's findings. The small sample size, with only nine participants per group, may have limited the statistical power to detect between-group differences. Because no formal a priori sample size calculation was performed, the study may not have been sufficiently powered to identify small or moderate differences between the training methods. Therefore, the non-significant ANCOVA result should be interpreted cautiously and cannot be taken as evidence that the two methods are equivalent. Future studies should recruit larger samples and use a formal a priori sample size calculation based on a clearly defined primary outcome and expected effect size.

Additionally, performance evaluation relied solely on shooting accuracy tests conducted under standardized testing conditions. Although the two training methods incorporate different movement sequences before shooting, the present study did not directly assess decision-making speed, performance under defensive pressure, or transfer to actual match situations. Therefore, the findings should be interpreted in relation to shooting accuracy rather than broader game performance. Future studies should incorporate representative game-based assessments to investigate whether these training methods also influence decision-making and shooting performance under match-like conditions.

Overall, both First-Touch Shooting and Shooting After Dribble groups showed improvements in shooting accuracy following the intervention. However, no statistically significant difference was found between the two training groups after adjusting for baseline shooting accuracy. Although Shooting After Dribble showed a numerically higher adjusted posttest mean and a larger within-group effect size, these findings should be interpreted cautiously given the small sample size. The similarity in the primary objective and some task demands of the two drills may provide a theoretical explanation for the observed pattern of results, but this interpretation requires further investigation. Future studies with larger samples and more rigorous designs are needed to clarify the comparative effects of these training methods.

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Conclusion

Both First-Touch Shooting and Shooting After Dribble training were followed by improvements in shooting accuracy. However, ANCOVA revealed no statistically significant between-group difference after adjustment for baseline shooting accuracy ($F = 0.994$, $p = 0.335$). Given the small sample size, the absence of a statistically significant difference should not be interpreted as evidence of equivalence between the two methods. Both methods may be considered options for shooting accuracy training in youth soccer players, although further research involving larger samples is needed to clarify their comparative effects.

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